JOHNS HOPKINS
WHITING SCHOOL of ENGINEERING

FPGA Design Using VHDL

Module 1A: Course Introduction

Course Prerequisites

- Required:
  - Basic Course in digital design
  - Understand combinational gates
  - Understand sequential gates
- Not required, but highly recommended:
  - Perform basic binary arithmetic
  - Convert between binary, decimal, hex

Course Description

- Hardware coding course to design hardware for FPGAs using VHDL
- Lab heavy course
  - Labs will increase in complexity and build upon each other
  - Important to start lab assignments early to have time to debug and work through issues that generally come up with FPGA design
- Weekly participation and quizzes
  - Make sure to participate early in the module week and come back near the end of the module to read/respond to other classmates
- One midterm exam

Course Objectives

- Translate a functional system description into appropriate digital blocks coded in VHDL.
- Perform synthesis, place-and-route, and map a digital design onto a target FPGA.
- Discuss the behavior and performance of designs targeting FPGAs.
- Implement Finite State Machines (FSMs) to solve digital design problems.
- Utilize the Nexy's development board to solve common design problems in the Xilinx Vivado environment.
- Simulate the behavior of VHDL designs using test benches written in VHDL and simulated using Vivado Simulator.

Course Grading

- Graded Weekly Quizzes – 10%
- FPGA Projects – 45%
- Midterm – 35%
- Weekly Discussions/Participation – 10%

Midterm Exam

- Worth 35% of final grade
- Material includes topics from lectures, quizzes, assigned readings, discussion forums, and lab assignments
- More details will be given prior to exam
- Knowledge of VHDL concepts, syntax, and behavior are required
- Exam assigned on weekend of Module 10

Miscellaneous Notes

- Lab assignments are to be completed independently
  - Copying assignments from other students or from the web is not allowed
- Assignments are to be well documented and appropriately designed

Miscellaneous Notes Continued

- Lab Assignment Grading
  - Meeting all requirements may not guarantee a complete grade
  - It is up to the instructor's discretion how points are awarded, or deducted, for each lab
- As an Example:
  - Bugs not specifically detailed in lab are still considered bugs
  - Copy/Paste utilization summary with no analysis is not acceptable (except for lab 1)

Discussion Forum

- For your benefit!
- Participation is an important part of this course and participation counts towards final grade!
- Keep in mind:
  - We learn only one thing from the correct solution, we learn **many** things from the incorrect solution

Late Submissions

- Individual Lab assignments will be accepted up to 1 week past the deadline with a penalty of 5 points taken off the grade
- Quizzes will not be accepted after the due date

Course Material

- Class Material:
  - Online
  - Notes, Assignments, reference manuals
- Textbook:
  - VHDL for Logic Synthesis, Andrew Rushton
- Development Board:
  - Nexy's A7 (or Nexy's 4 DDR) Board

Additional Hardware

- Certain Labs require extra hardware
  - VGA Monitor
    - If you don't have a VGA monitor but have an HDMI monitor, please email me for details on an adapter that can be used
  - Oscilloscope (Optional but highly recommended)
    - Can use sound recording application on phone as substitute

Teaching Style

- Course assumes:
  - No VHDL knowledge
  - No experience designing complex digital systems
  - Knowledge of AND, OR, MUX, Flip-flops, latches
- Taught with voice-over-powerpoint
- Assignment issues – reach out over email or office hours for advice on how to keep moving forward